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$$\int \frac{x^3 + x}{x^2 - 1} dx$$

$$\frac{x^3 + x}{x^2 - 1} = x + \frac{2x}{x^2 - 1}$$

$$x^2 - 1 = (x - 1)(x + 1)$$

$$\frac{2x}{x^2 - 1} = \frac{2x}{(x - 1)(x + 1)} = \frac{A}{x - 1} + \frac{B}{x + 1}$$

$$A = \left. \frac{2x}{x + 1} \right|_{x=1} = \frac{2}{2} = 1$$

$$B = \left. \frac{2x}{x - 1} \right|_{x=-1} = \frac{-2}{-2} = 1$$

$$\frac{x^3 + x}{x^2 - 1} = x + \frac{1}{x - 1} + \frac{1}{x + 1}$$

$$\int \frac{x^3 + x}{x^2 - 1} dx = \int x dx + \int \frac{dx}{x - 1} + \int \frac{dx}{x + 1}$$

$$= \frac{x^2}{2} + \ln|x - 1| + \ln|x + 1| + c$$

References